Power BI Development Track

Menofia Branch

2025/2026

MASSAR

Smart Transport Intelligence Platform

Graduation Project Documentation

Team Members:

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1. Introduction

Massar is a multi-domain smart transportation platform designed for the Egyptian market. It serves three business domains under a single platform**: Mobility** (passenger rides), **Carrier** (parcel and light delivery), and **Freight** (heavier cargo). The platform generates large volumes of operational data, including rides, payments, driver sessions, subscriptions, promotions, and ratings.

The goal of this project is to build a complete, end-to-end data and AI solution on top of that operational data: a clean operational database (OLTP), a dimensional data warehouse, an automated transformation pipeline using dbt, a set of Power BI dashboards published on Microsoft Fabric, and an AI assistant that answers business questions in natural language.

1.1 Business Domains

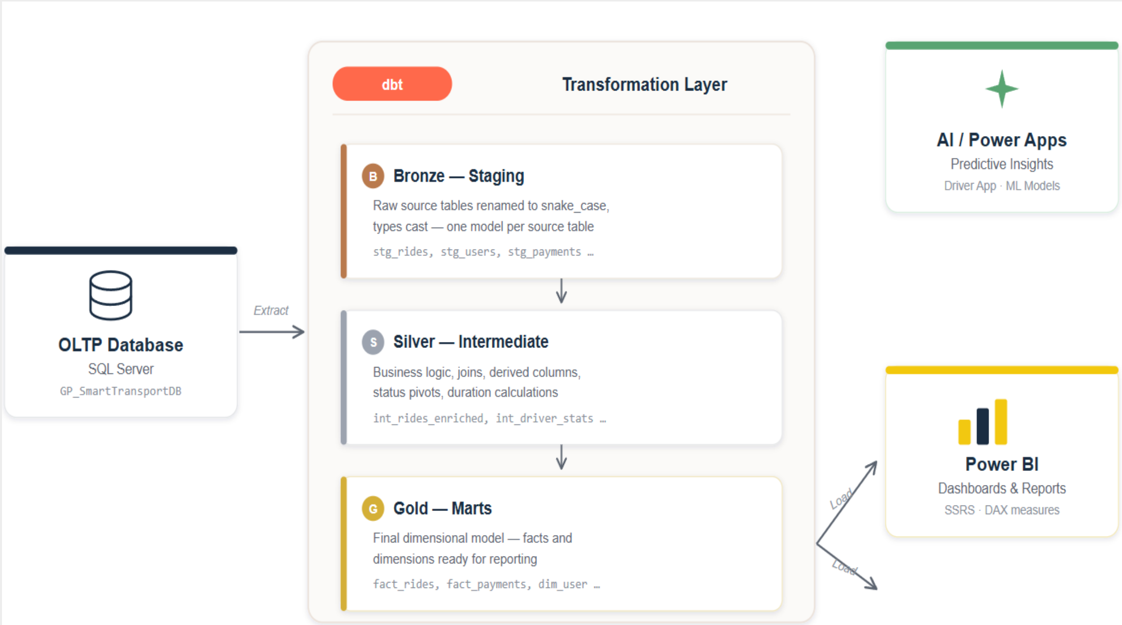
Domain	Description
Mobility	On-demand passenger rides for individuals
Carrier	Parcel and light-goods delivery services
Freight	Larger cargo and freight transport

1.2 Objectives

- Design and implement a normalized OLTP database for daily operations.
- Build a Kimball dimensional data warehouse for analytics and KPIs.
- Automate transformations with dbt using a layered (medallion) architecture.
- Deliver Power BI dashboards published on Microsoft Fabric.
- Add an AI assistant (LLM + RAG + text-to-SQL) over a governed semantic layer.

1.3 Solution Overview

Data flows from the operational database into the warehouse through dbt (Bronze -> Silver -> Gold). The Gold tables feed Power BI for fixed dashboards, while a separate governed semantic view layer feeds the AI assistant for free natural-language questions. This keeps reporting and AI cleanly separated.



2. Database Design

The operational database (OLTP) is implemented on Microsoft SQL Server as GP_SmartTransportDB. The design started from a conceptual model (ERD) and was then translated into a normalized physical schema.

2.1 Conceptual Design (ERD)

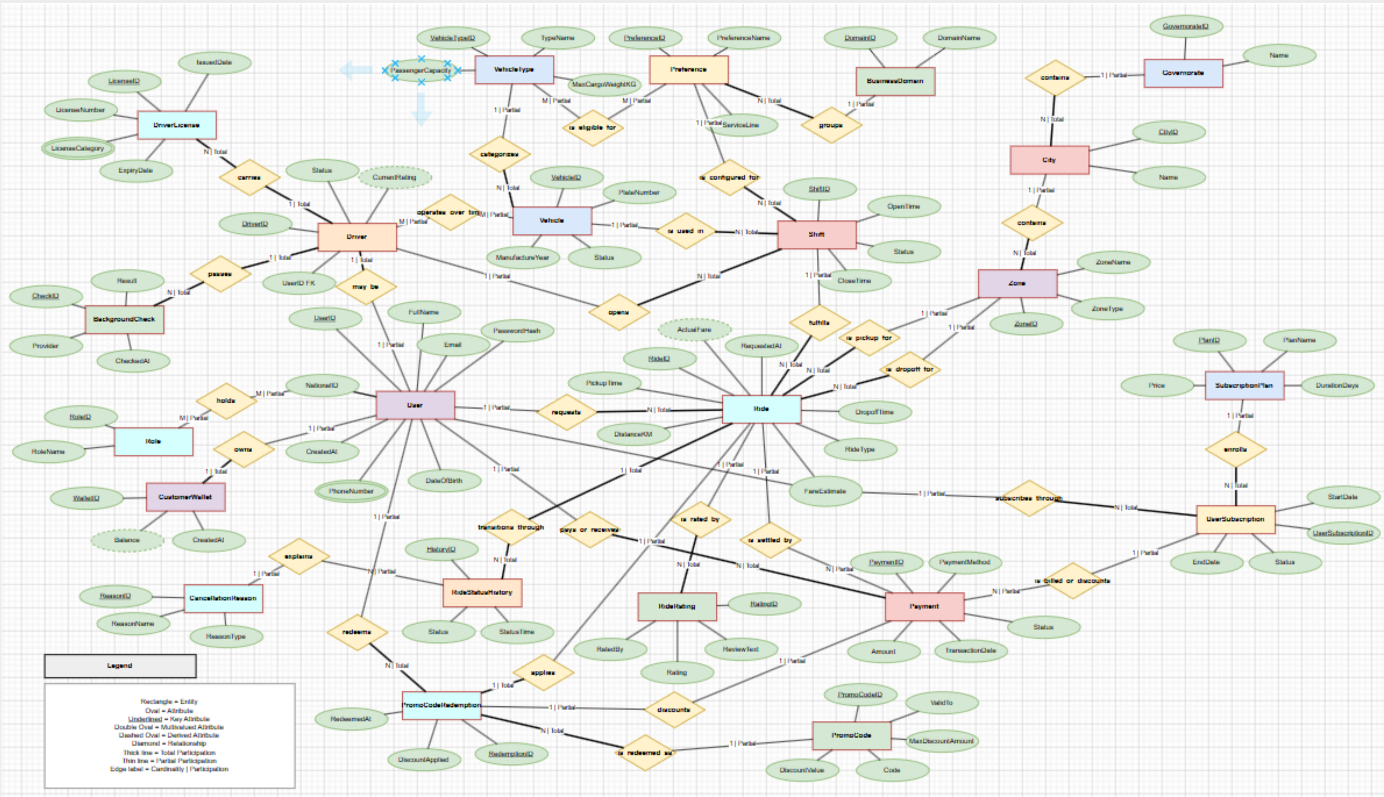
The conceptual model describes the business entities and their relationships independently of any technology. The main entity groups are:

- Users and Identity: User, Role, Driver, Wallet, Driver License, Background Check.
- Ride Operations: Ride, Ride Status History, Ride Rating, Session (shift), Cancellation Reason.
- Geography: Governorate -> City -> Zone hierarchy.
- Vehicles and Preferences: Vehicle, Vehicle Type, Driver-Vehicle, Preference, Business Domain.
- Financials: Payment, Promo Code, Promo Code Redemption, Subscription Plan, User Subscription.

Key Relationships

Relationship	Cardinality	Description
User - Driver	1:1 (optional)	A driver is a specialization of a user
Driver - Vehicle	M:N	Linked through DriverVehicle
User - Ride	1:N	A user requests many rides
Session - Ride	1:N	A driver fulfills rides within a session
Ride - Payment	1:1 / 1:N	Payment settles a ride
Ride - Zone	N:1 (x2)	Pickup zone and dropoff zone

Ride - RideStatusHistory	1:N	Status timeline of a ride
User - UserSubscription	1:N	Subscription enrollment



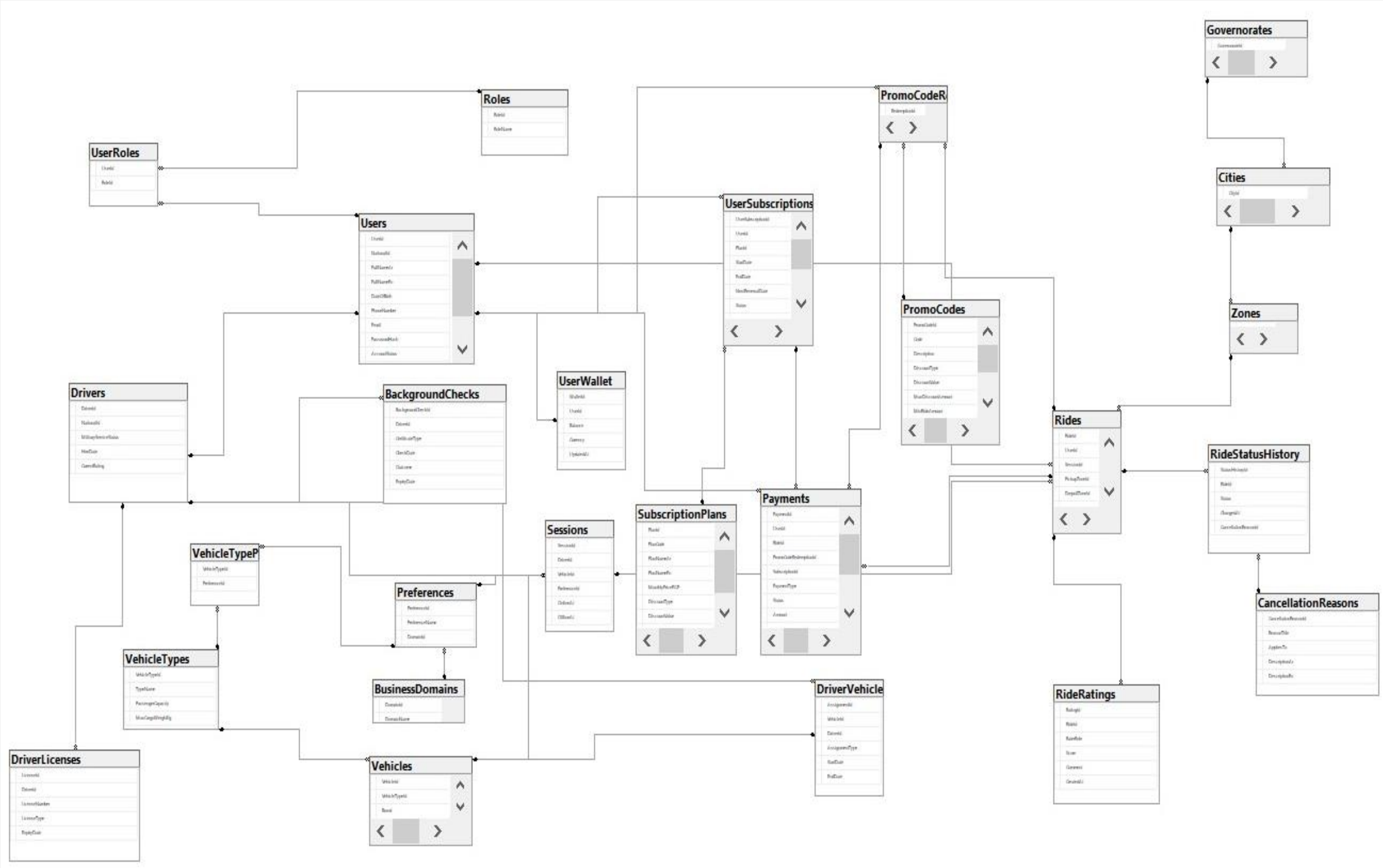
2.2 Schema Design (Physical)

Main Tables

Group	Tables
Lookup / Reference	Roles, BusinessDomains, VehicleTypes, Governorates, Cities, Zones, CancellationReasons, SubscriptionPlans, Preferences
Users and Drivers	Users, UserRoles, UserWallet, Drivers, DriverLicenses, BackgroundChecks
Fleet	Vehicles, DriverVehicle, VehicleTypePreference
Operations	Sessions, Rides, RideStatusHistory, RideRatings
Financials	Payments, PaymentMethods, PromoCodes, PromoCodeRedemptions, UserSubscriptions

Constraints and Integrity

- Primary keys on every table.
- Foreign keys enforcing all relationships.
- CHECK constraints on enumerations (roles, business domains, vehicle types).
- Validation scripts for business rules (e.g. rating direction passenger/driver).


Governorates
Roles
PromoCodeR
UserRoles
Cities
Users
UserSubscriptions
PromoCodes
Zones
Drivers
BackgroundChecks
UserWallet
Rides
RideStatusHistory
Payments
SubscriptionPlans
Sessions
VehicleTypeP
Preferences
CancellationReasons
VehicleTypes
BusinessDomains
DriverVehicle
RideRatings
DriverLicenses
Vehicles

3. Data Warehouse (dbt) and Semantic Layer

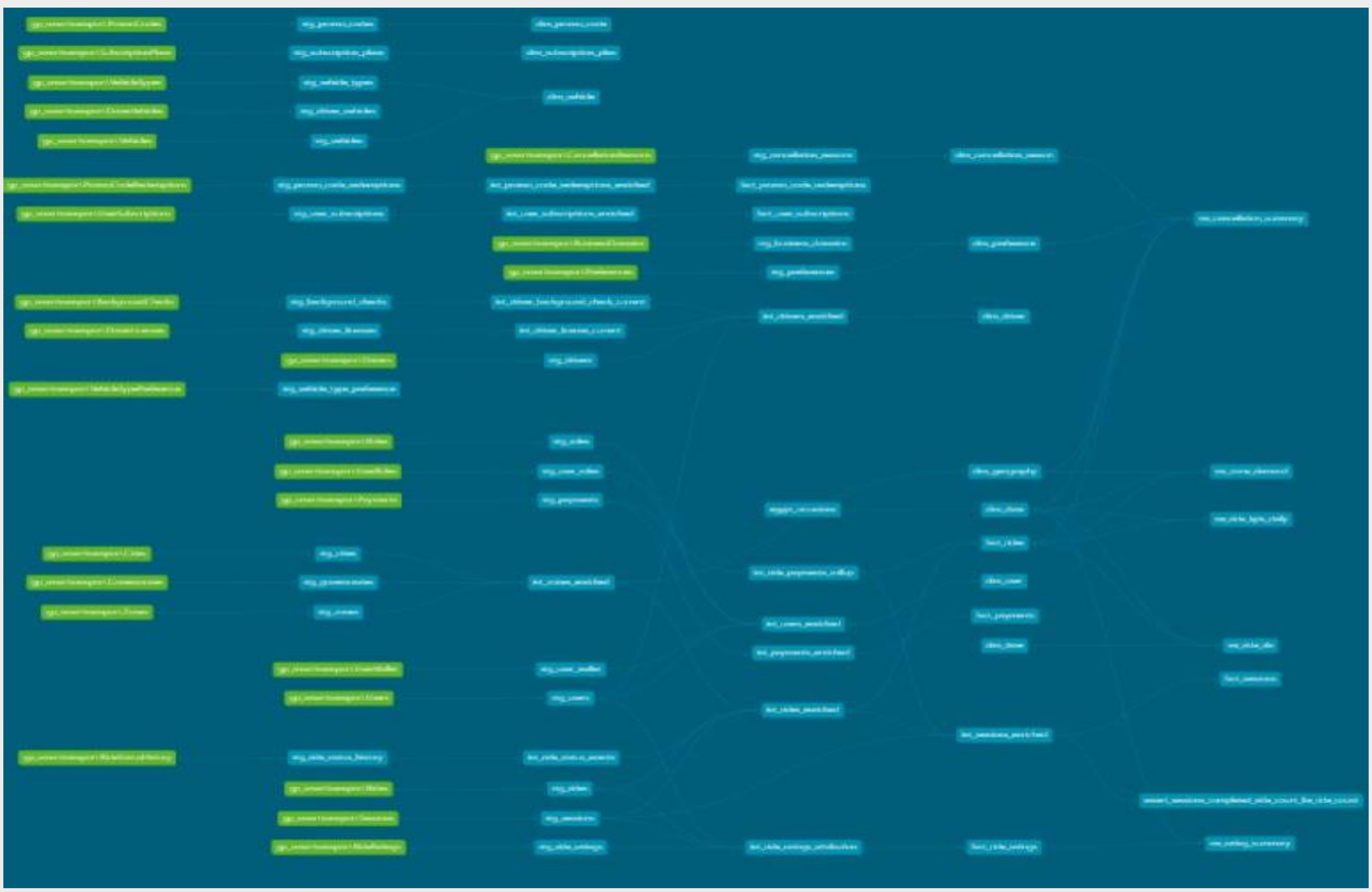
The data warehouse (GP_SmartTransportDWH) follows Kimball dimensional modeling (galaxy schema). All transformations are built and tested with dbt in the massar_analytics project, organized into a layered architecture.

3.1 dbt Layers (Medallion Architecture)

Layer	Folder	Schema	Material.	Purpose
Bronze (Staging)	models/bronze/	staging	View	1:1 mirrors of OLTP: cleaning, renaming, typing (stg_*)
Silver (Intermediate)	models/silver/	intermediate	View	Joins, enrichment, business logic (int_*)
Gold (Marts)	models/gold/	marts	Table	Star schema: dimensions and facts for Power BI
Gold (Semantic)	models/gold/semantic/	semantic	View	Governed KPI views used by the AI only (vw_*)

Lineage Flow

OLTP sources -> stg_* (Bronze) -> int_* (Silver) -> dim_* and fact_* (Gold/marts) -> Power BI, and in parallel -> semantic vw_* (Gold/semantic) -> AI assistant.



3.2 Gold Layer - Dimensions and Facts

Fact Tables (6)

Fact	Grain	Description
fact_rides	One row per ride	Core ride fact; accumulating snapshot for status timestamps
fact_payments	One row per payment	Revenue, refunds, commissions
fact_sessions	One row per session	Driver shift metrics
fact_ride_ratings	One row per rating	Role-playing dim_user (rater/rated)
fact_user_subscriptions	One row per subscription	Subscription lifecycle
fact_promo_code_redemptions	One row per redemption	Promo usage

Dimension Tables (10)

- dim_date - calendar spine; Egypt weekend (Fri-Sat); occasions (Ramadan, Eid).
- dim_time - minute grain (1440 rows); period of day; peak-hour flag.
- dim_user, dim_driver, dim_vehicle.
- dim_geography - flattened Governorate -> City -> Zone.
- dim_preference - preference and business domain.
- dim_cancellation_reason - reason and applies_to.
- dim_subscription_plan, dim_promo_code.

3.3 Semantic Layer for AI

On top of the Gold mart tables, a dedicated semantic layer exposes governed, pre-aggregated KPI views (schema: semantic). These views contain no PII and are the only objects the AI assistant is allowed to query. Power BI keeps using the mart tables; the AI uses the semantic views - a clear security boundary.

Semantic View	Purpose
vw_ride_kpis_daily	Daily rides, revenue, completion and cancellation
vw_cancellation_summary	Cancellations by reason, zone, and domain
vw_zone_demand	Pickup-geography demand and revenue
vw_ride_sla	Peak hours and accept/complete SLA
vw_rating_summary	Ratings by role and score (no PII)

dbt Quality and Tests

- Tests: unique, not_null, relationships, accepted_values.
- dbt_utils.unique_combination_of_columns for composite keys.

- egypt_occasions seed feeds dim_date (Ramadan / Eid al-Fitr / Eid al-Adha).
- Lineage and docs via dbt docs generate / dbt docs serve.

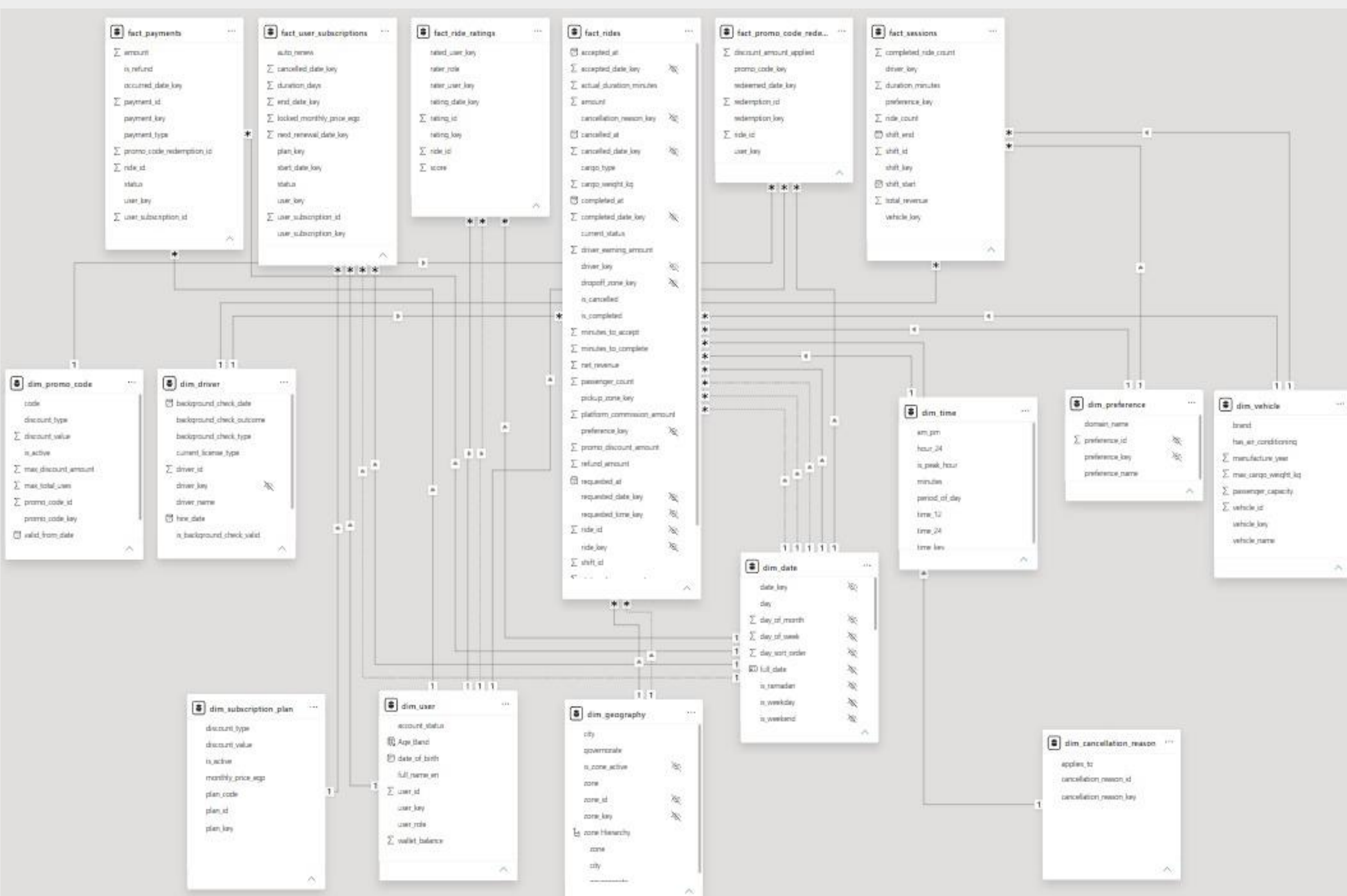
4. Power BI Dashboards and Semantic Model

On top of the data warehouse we built a Power BI semantic (data) model and a set of interactive dashboards, then published them to Microsoft Fabric.

Property	Value
Tool	Microsoft Power BI
Publishing platform	Microsoft Fabric
Workspace	Massar_Project
Report / Semantic model	Massar_BI_Project
Data source	GP_SmartTransportDWH (schema: marts)
Number of dashboard pages	21+ pages

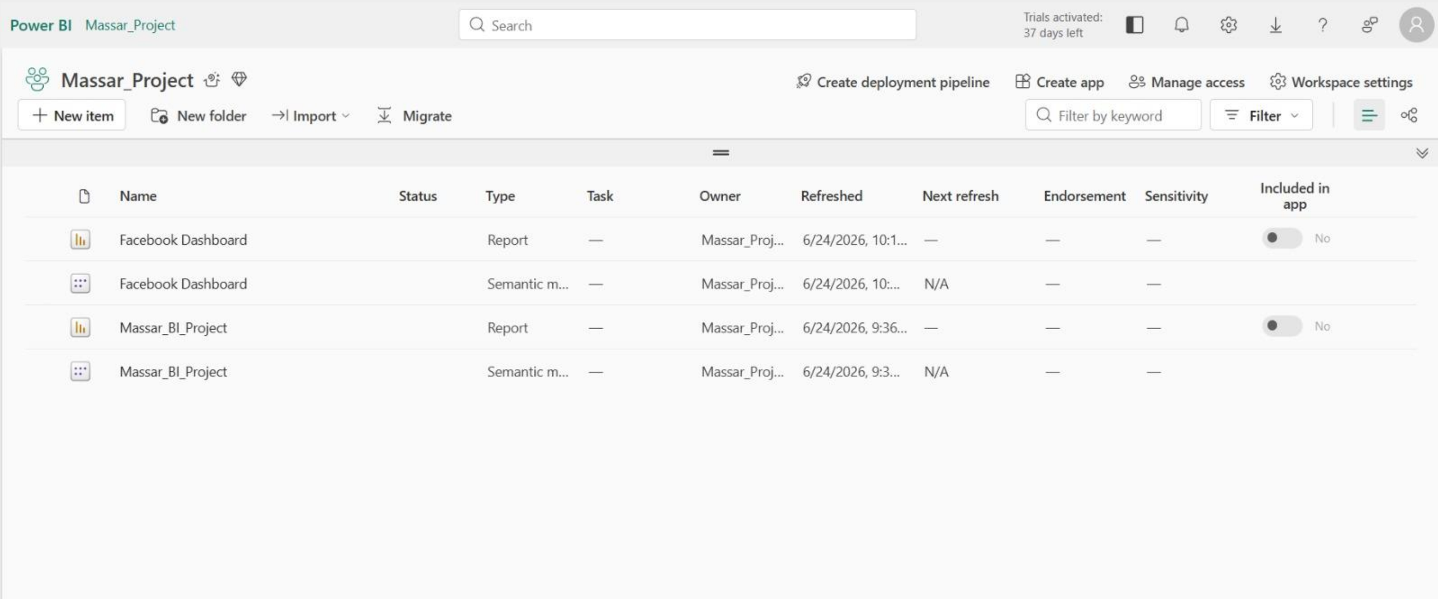
4.1 Power BI Semantic Model (based on the DWH)

The Power BI semantic model is built directly on the warehouse Gold layer. It imports the 6 fact tables and 10 dimensions and defines the relationships of a star schema: one-to-many from each dimension to the facts. dim_date is marked as the date table on **full_date**. Alternate date roles (requested, accepted, completed, cancelled) are modeled as inactive relationships and activated in DAX using USERELATIONSHIP. Fact-to-fact links are kept inactive; facts relate only through shared dimensions.

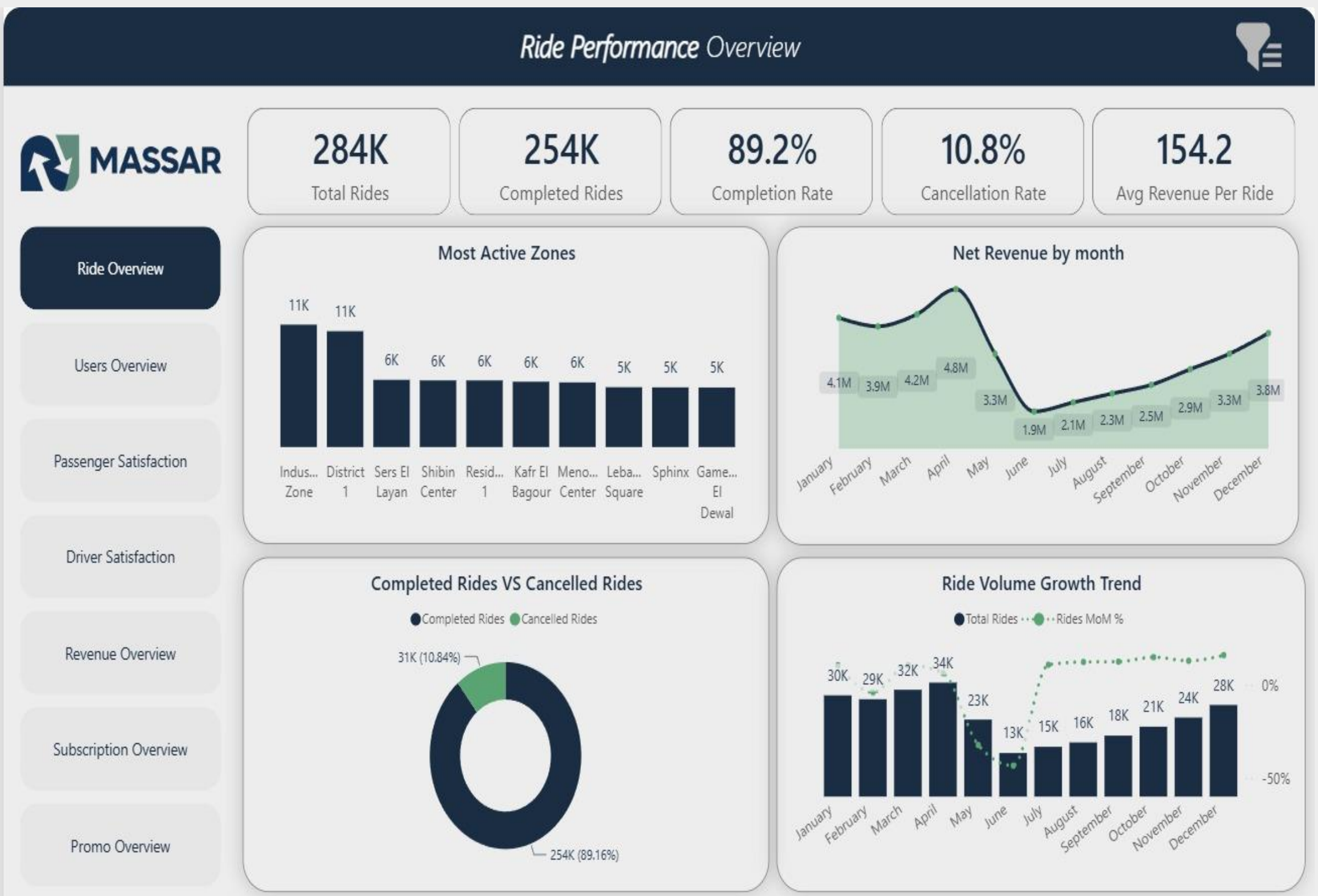


4.2 Power BI Dashboards

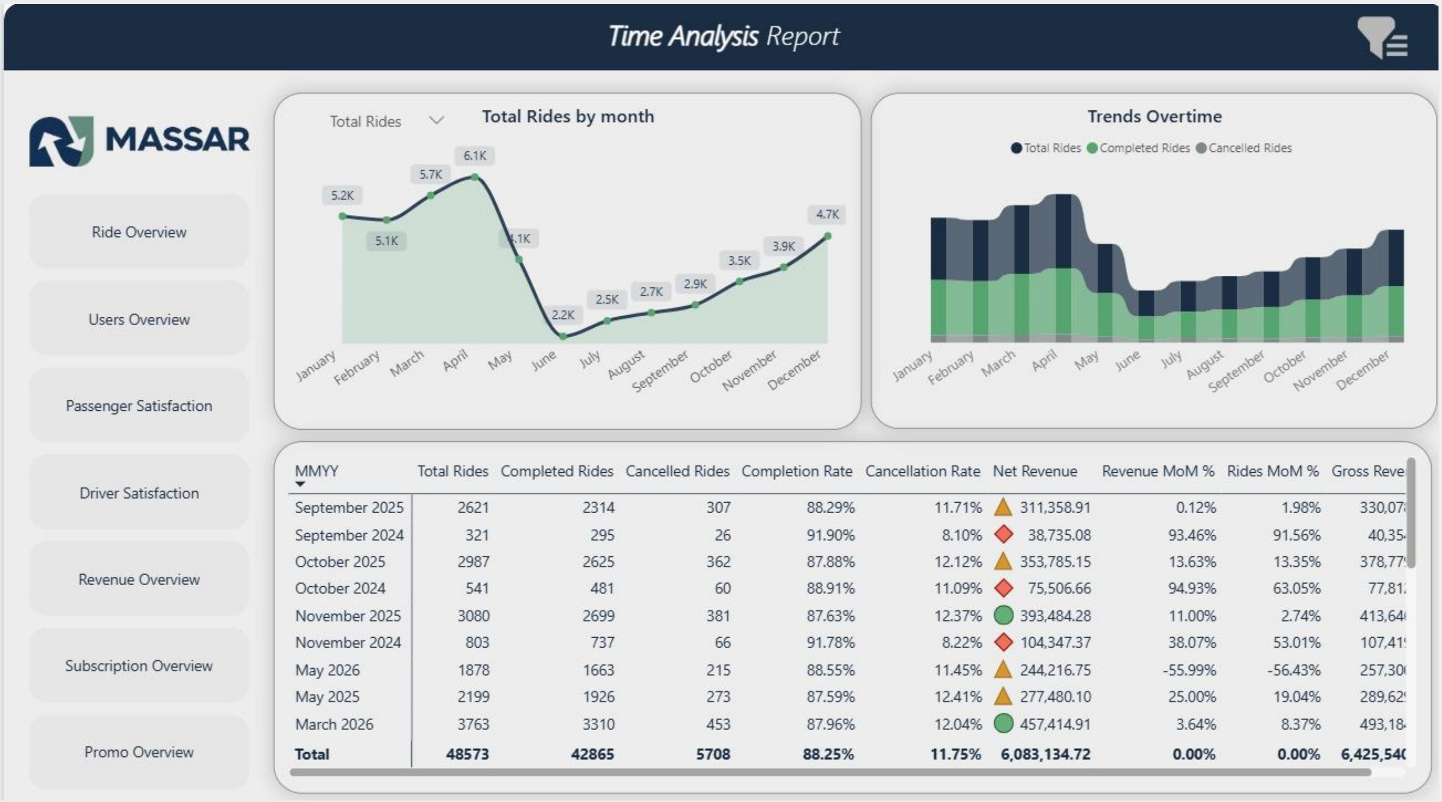
Microsoft Fabric workspace (Massar_Project).



Ride Overview



Time Analysis



MMYY	Total Rides	Completed Rides	Cancelled Rides	Completion Rate	Cancellation Rate	Net Revenue	Revenue MoM %	Rides MoM %	Gross Reve
September 2025	2621	2314	307	88.29%	11.71%	311,358.91	0.12%	1.98%	330,07
September 2024	321	295	26	91.90%	8.10%	38,735.08	93.46%	91.56%	40,35
October 2025	2987	2625	362	87.88%	12.12%	353,785.15	13.63%	13.35%	378,77
October 2024	541	481	60	88.91%	11.09%	75,506.66	94.93%	63.05%	77,81
November 2025	3080	2699	381	87.63%	12.37%	393,484.28	11.00%	2.74%	413,64
November 2024	803	737	66	91.78%	8.22%	104,347.37	38.07%	53.01%	107,41
May 2026	1878	1663	215	88.55%	11.45%	244,216.75	-55.99%	-56.43%	257,30
May 2025	2199	1926	273	87.59%	12.41%	277,480.10	25.00%	19.04%	289,62
March 2026	3763	3310	453	87.96%	12.04%	457,414.91	3.64%	8.37%	493,18
Total	**48573**	**42865**	**5708**	**88.25%**	**11.75%**	**6,083,134.72**	**0.00%**	**0.00%**	**6,425,54**

5. SSRS Reports

#	Report	Key Content
1	Executive Overview	Total rides, completion rate, net revenue, avg accept time, revenue by month
2	Financial Report	Gross/net revenue, driver earnings, commission, discounts, refunds
3	Ride Operations	Rides by status, completion by period of day, top cancellation reasons, SLA
4	Driver Performance	Sessions, online minutes, earnings, compliance rate, top drivers
5	User and Subscription	Plans (Basic/Plus/Premium), monthly active riders, role distribution
6	Promo and Ratings	Promo ROI, rating distribution, average rating by governorate

1.

Start Date 2025-06-01 End Date 2026-06-01 View R

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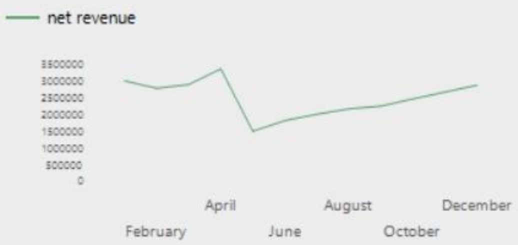
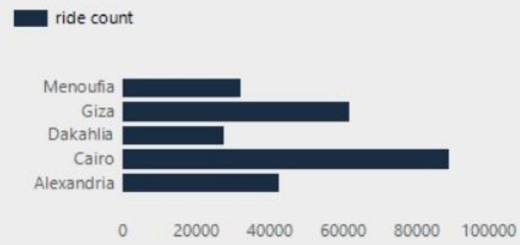
Executive Overview Dashboard

Total Rides	Completion Rate	Net Revenue	Avg Accept Time
219249	89.000000000000%	30,043,799.2EGP	1.12

Revenue by Month



Rides by Governorate



2.

Start Date 2026-01-01 End Date 2026-06-01

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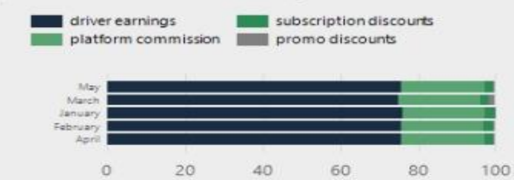
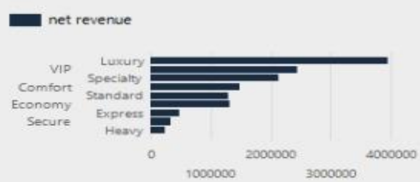
Financial Report

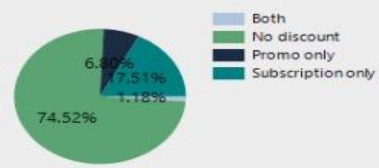
revenue split



Revenue by preference



Discount type distribution



year	month name	gross revenue	net revenue	driver earnings	platform commission	subscription discounts	promo discounts	refunds
2026	January	3,193.24EGP	3,024.63EGP	2,423.39EGP	683.52EGP	72.56EGP	10.48EGP	85.58EGP
2026	February	2,984.12EGP	2,803.85EGP	2,258.99EGP	637.15EGP	73.04EGP	21.41EGP	85.81EGP
2026	March	3,129.66EGP	2,918.89EGP	2,375.31EGP	669.96EGP	74.46EGP	49.60EGP	86.71EGP
2026	April	3,574.19EGP	3,384.03EGP	2,706.73EGP	763.44EGP	76.56EGP	22.69EGP	90.91EGP
2026	May	1,591.44EGP	1,518.41EGP	1,182.07EGP	333.41EGP	34.82EGP	11.04EGP	27.17EGP
Totals		14,472.66EGP	13,649.82EGP	10,946.50EGP	3,087.47EGP	331.45EGP	115.21EGP	376.19EGP

3.

Start Date 2026-01-01 End Date 2026-06-01 Governorate Cairo View Report

1 of 2 ? 100% Find | Next

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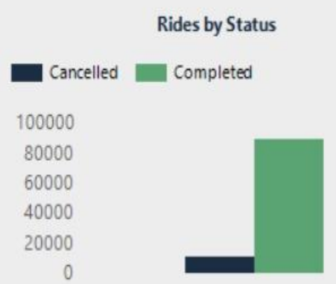
Ride Operations Report

governorate name en	city name en	avg accept minutes
Cairo	Downtown	1
	Heliopolis	1
	Maadi	1
	Nasr City	1
	New Cairo	1
	Shubra	1



6. Artificial Intelligence (AI Assistant)

Massar AI is an analytics assistant that lets users ask business questions in plain English. It combines Retrieval-Augmented Generation (RAG) with governed text-to-SQL over the semantic KPI views, and presents answers - with optional charts - through a chat interface. It complements Power BI; it does not replace it.

Component	Technology
User interface	Streamlit chat app
Orchestration	LangChain
LLM	Azure OpenAI (gpt-4.1-mini)
Embeddings	text-embedding-3-small
Vector database	ChromaDB (local)
Database access	SQL Server (schema: semantic) via pyodbc, read-only

6.1 How It Works

1. The user asks a question in the Streamlit chat.
2. Conversation memory resolves follow-up questions.
3. RAG retrieves relevant KPI/schema docs from Chroma.
4. The request is routed: definition vs KPI value vs analytics query.
5. SQL is generated (template or LLM) against the semantic views.
6. sql_guard validates it (SELECT-only, whitelisted views, no DDL/DML).
7. The query runs and the result is formatted (KPI % aligned with Power BI).
8. Agentic insights (e.g. month-over-month change) and a chart are added.

6.2 RAG Knowledge Base

- kpi_glossary.md - definitions of every KPI.
- semantic_schema.md - description of the semantic views.
- business_rules.md - domain and business logic.
- example_questions.md - sample questions and expected behavior.

6.3 Security (sql_guard)

- SELECT statements only - no INSERT/UPDATE/DELETE/DROP/EXEC.
- Whitelist limited to the 5 semantic views (semantic.vw_*).
- Row limit per query to protect the database.
- No PII anywhere in the semantic layer or RAG documents.

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AI Analytics Assistant

Agentic LangChain RAG + governed text-to-SQL over semantic KPI views

Top 5 pickup zones by ride count

The top 5 pickup zones by ride count are:

1. Sers El Layan in Menouf with 6,119 rides
2. Shibin Center in Shibin El Kom with 6,069 rides
3. Residential 1 in Sadat City with 6,067 rides
4. Industrial Zone in Sadat City with 5,987 rides
5. Kafr El Bagour in Shibin El Kom with 5,949 rides

📊 The top pickup zone, Sers El Layan in Menouf, had slightly more rides (6119) than the next four zones, which all had ride counts close to 6000, indicating a relatively even distribution among the top five zones.

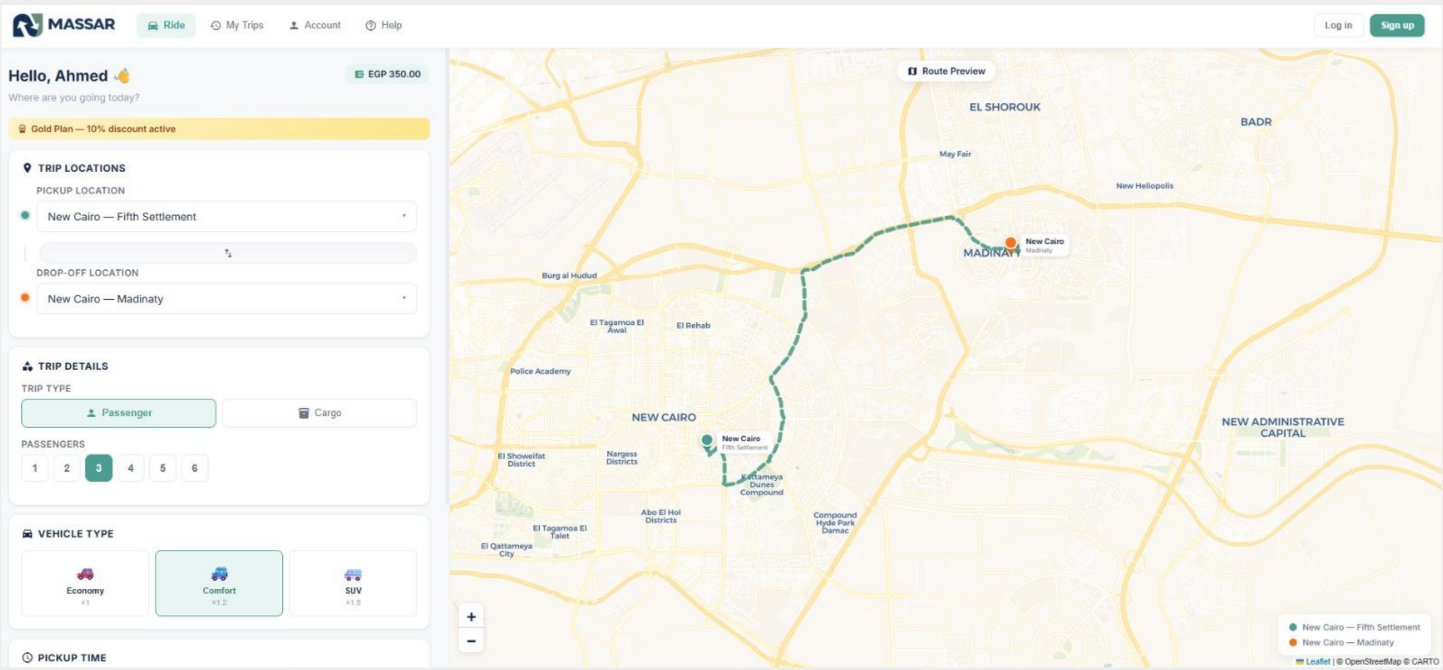
Ask about rides, KPIs, zones, SLA, or ratings...

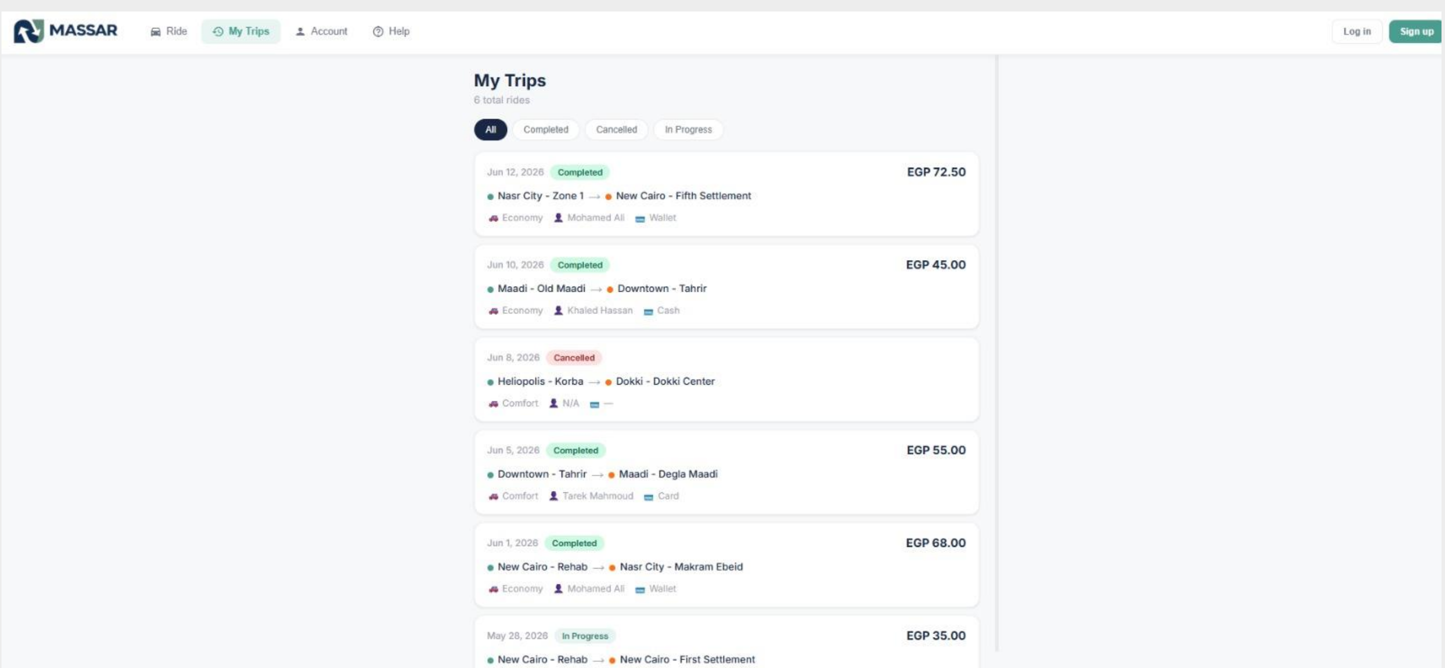
7. Application

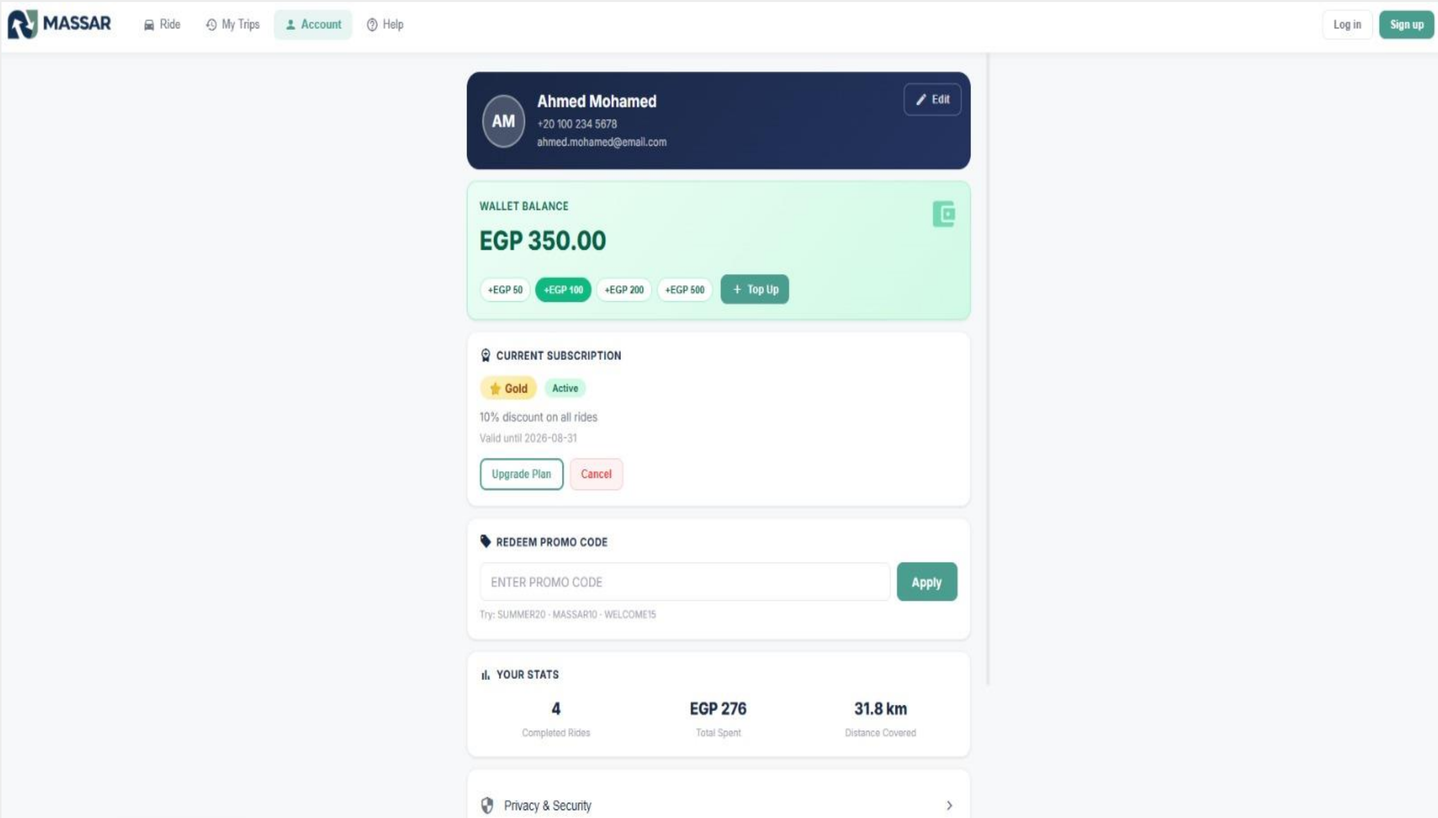
The application layer ties the platform together for end users. It provides the transport experience that produces the operational data and the analytical surfaces (dashboards and AI) that consume it.

7.1 Application Components

Component	Role
Transport apps (rider/driver)	Generate operational data (rides, payments, sessions)
Power BI / Fabric	Fixed analytical dashboards for stakeholders
Massar AI (Streamlit)	Conversational analytics over the semantic layer



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Ride
My Trips
Account
Help
Log in
Sign up
AM
Ahmed Mohamed
+20 100 234 5678
ahmed.mohamed@email.com
Edit
WALLET BALANCE
EGP 350.00
+EGP 50
+EGP 100
+EGP 200
+EGP 500
+ Top Up
CURRENT SUBSCRIPTION
Gold
Active
10% discount on all rides
Valid until 2026-08-31
Upgrade Plan
Cancel
REDEEM PROMO CODE
ENTER PROMO CODE
Apply
Try: SUMMER20 · MASSAR10 · WELCOME15
YOUR STATS
4
Completed Rides
EGP 276
Total Spent
31.8 km
Distance Covered
Privacy & Security

8. Tools and Technologies

Category	Tool / Technology
OLTP and DWH database	Microsoft SQL Server
Transformation	dbt Core + dbt-sqlserver, dbt_utils
Business intelligence	Microsoft Power BI
Cloud / publishing	Microsoft Fabric
AI - LLM	Azure OpenAI (gpt-4.1-mini)
AI - framework	LangChain
AI - vector DB	ChromaDB
AI - UI	Streamlit
AI - DB driver	pyodbc
Languages	Python, SQL, DAX
Version control	Git + GitHub (GPSmartTrasportation)
Modeling approach	Kimball dimensional modeling